

DL Exam Topics

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Foreword

If you pass the 'Basic Course Concepts' test (by answering at least half of the questions correctly), you will be randomly assigned one of the questions listed below. You can earn up to 4 additional points for this part. Correct answers for this part are not provided, you can write them down yourself. Your answers to these questions will demonstrate a deep understanding of the course topics, rather than being a collection of answers to the 'Basic Course Concepts' questions. Prepare a pen and a sheet of paper; it may be easier to draw pictures to answer the questions. The examiner may ask additional questions within the scope of the topic.

To receive a grade of 9 or 10, students may be asked additional questions beyond the initial topic. These may be questions from the list below, as well as tasks/questions 'to think about' from the examiner. They may not be included in the list below. It is guaranteed that additional questions will be on topics that were covered in the course.

The second part of the exam lasts up to 12 minutes in total.

Topics

1. Linear layer

What is a linear (fully connected) layer? Write down the formula. Formulate the Universal Approximation Theorem. Why is nonlinearity between linear layers necessary? What types of nonlinearity are you familiar with? What are their advantages and disadvantages? How is the number of trainable parameters in a linear layer calculated? How does backpropagation work? Show this using the example of a small fully connected neural network, writing out the formulas.

2. Gradient descent and loss-functions

Describe the main types of gradient descent (GD, SGD, mini-batch). What are Adam and AdamW? What loss functions are used for classification and regression tasks? How does the choice of lr and the use of lr-scheduler affect the training process?

3. Initialization

Why is correct weight initialization important in neural networks? Describe common initialization methods (Xavier, He) and explain their principles. How can incorrect initialization lead to gradient vanishing or exploding? Is it possible to initialize weights with a constant? What will happen then?

4. Dropout, BatchNorm

Explain how Dropout works in train and eval modes. What is BatchNorm and how does it affect the training process? How does it behave in train and eval modes? Is it necessary to normalize data before feeding it into a neural network? Does this apply only to fully connected networks or is it also applicable to CNNs?

5. Convolution in CNN

Write down the convolution layer formula, indicate how the number of parameters is calculated (with and without bias). How is the number of multiplication operations calculated during convolution for given input and output tensor sizes? What other parameters of convolution layers do you know, and how do they affect performance? What is a perception field? How is it calculated?

6. Convolutional neural network architectures

What well-known CNN architectures do you know (AlexNet, VGG, ResNet, Inception)? You can draw a diagram. What are the advantages and disadvantages of different architectures? Which architecture would you prefer to use now?

7. Transfer learning and fine-tuning in CNN

What is transfer learning and how can it be applied to computer vision tasks? Explain the concept of fine-tuning and give an example of its use. What features should be considered when adapting a pre-trained model to a new task? What does it mean to 'freeze' the weights of a neural network?

8. Pooling and padding

What types of pooling are you familiar with? Why do we need pooling? What types of padding exist and how do they affect the size of the output tensor? Give examples where these techniques are used to improve network architecture.

9. Data augmentation in computer vision tasks

What role does data augmentation play in CNN training? What types of augmentation are most common, and in what areas might their use be limited? Give examples and discuss the pros and cons of this approach.

10. Detection

Explain the difference between classification, segmentation, and detection tasks in computer vision. What is mAP and how is it used to evaluate detection quality? Discuss the development of architectures for detection tasks and give an example of a model.

11. Segmentation

Explain the difference between classification, segmentation, and detection tasks in computer vision. What is IoU and how is it used to evaluate segmentation quality? Discuss the differences between semantic and instance segmentation and give an example of a model.

12. Identification

Explain the difference between classification and identification tasks in computer vision. Why should identification tasks not be solved using classification? What loss functions do you know for this task? How are models for solving this type of task structured?

13. Recurrent Neural Networks (RNN)

What are RNNs and how do they work? What is the motivation for using RNNs to work with sequential data? What problems arise when training RNNs (gradient vanishing/explosion) and how can they be solved? Give examples of using RNNs in classification and regression tasks on text data.

14. LSTM and GRU

Explain the weaknesses of standard RNNs and how LSTM/GRU solve these problems. Describe the structure of LSTM and its main components (memory cell, input/output and forget gates). Compare LSTM and GRU. What are the main differences? Write down the formulas for these layers or at least name the main components and compare the number of trainable parameters.

15. Embeddings for text data

What are embeddings and why are they needed in text processing? Describe the word2vec model, its advantages and disadvantages. How do embeddings affect the quality of natural language processing? How to train your embeddings in PyTorch?

16. Generative Adversarial Networks (GAN)

What is a GAN and what components does this architecture consist of? Describe the role of each component and formulate the task of training a GAN as a minimax optimization problem. What are the main difficulties encountered when training a GAN? In what cases is the GAN model more suitable than other generative models?

17. **Diffusion Models**

Describe the general idea of diffusion models. How are the forward and backward processes formulated? Why does training diffusion models boil down to the task of predicting noise? What are the main advantages of diffusion models? What are the disadvantages of this approach? Why do we need a scheduler?

18. **Autoencoder and Variational Autoencoder (AE/VAE)**

What is an autoencoder and what tasks does it solve? What is the fundamental difference between a variational autoencoder and a regular autoencoder? What role does KL divergence play in the VAE loss function? What are the advantages and disadvantages of this model? What problems can be solved with such models?

19. **Normalizing Flows (NF)** What are normalizing flows and what is the idea behind this approach? How is the variable substitution formula used to calculate likelihood? What requirements are imposed on transformations in normalizing flows? Describe (formulas are optional) RealNVP. Compare normalizing flows with other models.

20. **Attention Mechanism**

What is the attention mechanism and what problem does it solve in models for sequential data? Describe the general idea of query-key-value. How does attention affect the interpretability of models? Give examples of tasks where the use of attention provides a significant advantage over models without attention.

21. **Transformers** Describe the Transformer architecture. What is the difference between self-attention and cross-attention? Why does Transformer not use recurrent connections? How is token order information encoded? Give examples of tasks where Transformers perform best. How is this architecture better than RNN?

22. **Low-Rank Adaptation (LoRA)**

What is LoRA and what tasks is it used for? What is the idea behind low-rank approximation when retraining large language models? What advantages does LoRA have over full fine-tuning and adapter-based approaches? How does LoRA affect the number of trainable parameters and memory consumption?